Scaling Up secure Processing, Anonymization and generation of Health Data for EU cross border collaborative research and Innovation

SECURED
EU Project

D2.3 — Annex: Synthetic data generation for UC2 - Time series oversampling generation

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1 Introduction

Hypotension is a medical condition where the mean blood pressure (MBP) drops, typically, under 65 mmHg for a minute to few minutes, as reported in [1]. This condition is generally considered low blood pressure. This health condition is evident when patients shift from sitting to a standing position, called "orthostatic hypotension” (OH), when patients exhibit a permanently lower figure, also called "constitutional hypotension" (CH) or during surgical operations. Despite its historical marginality, hypotension is gaining more relevance in the current decades, with an increase in hypotension cases in the population. In most of the occurrences, it is affecting old adults, young women with high heart rate, patients with cardiovascular problems and Intensive Care Unit (ICU) patients [2]. The causes behind hypotension events are related but not limited to: ageing, diabetes, autoimmune system, and taking medicines that cause side effects.

While hypotension is clinically recognised as a frequent complication in this patient population, these episodes are often temporary, typically lasting only a few minutes. From a monitoring perspective, this creates a significant temporal imbalance; the overall duration of hypotensive events represents a small fraction of the total observation period. Consequently, when the monitoring data is partitioned into discrete time windows, the subset of intervals exhibiting hypotension is vastly outnumbered by those reflecting haemodynamic stability. In other words, among a patient's consecutive time windows, those showing hypotension constitute only a small subset, making hypotension a rare event in statistical terms. This alternative point of view reveals the time-scarcity nature of hypotension, posing a hard challenge for event detection tasks.

This study explores the use of Machine Learning to design a synthetic time-series generator that replicates the complex patterns of MBP preceding a hypotension event. The goal is to artificially expand our dataset, balancing the ratio of MBP windows that prelude hypotensive intervals against stable ones. The augmented dataset is used as input to a Time Series Classifier represented by a Long-Short Term Memory (LSTM) Deep Learning architecture [3], equipped with a final density layer for classification [1].

Balancing the ratio between windows of MBP leading to hypotension and haemodynamically stable windows ensures that the predictive model is trained on a more even distribution of data, rather than being biased toward the much more frequent periods of normal blood pressure.

One effective strategy to mitigate the scarcity of event data is the implementation of a generator designed to oversample the minority class, specifically the sequences containing hypotensive events. This approach focuses on the synthesis of plausible new sequences derived from actual clinical data where a rapid, unexpected decline in blood pressure was observed.

Historically, oversampling was often achieved through simple random oversampling, which involves duplicating existing minority samples; here we call this method Random Oversampling, or just "rover" for brevity. However, this method frequently leads to overfitting, as the model essentially memorises specific instances rather than learning the underlying physiological patterns. To overcome this limitation, more sophisticated techniques have been developed, most notably the Synthetic Minority Over-sampling Technique (SMOTE) [4]. Oversampling the minority class provides a more diverse and representative training set, allowing Artificial Intelligence (AI) to better generalise the arrival of a hypotensive crash.

The sudden onset of hypotension can be a destabilising element in a patient's daily life, creating a sense of vulnerability and a fraught relationship with their own health. These unexpected drops in blood pressure become far more perilous during surgical procedures, where they can compromise the success of the operation and patients‘safety. If medical professionals could accurately forecast the onset of a hypotensive event, the entire framework of care would shift. Such foresight would allow nurses and doctors to take pre-emptive measures, tailor anaesthetic management more precisely, and provide a critical safety net that could save lives in both routine and emergency settings.

This study presents a synthetic data generator that replicates the complex morphology of hypotension within Mean Blood Pressure (MBP) time series. By generating realistic synthetic sequences, the framework addresses data scarcity and bolsters the efficacy of machine learning models in detecting clinical hypotension. Extensive

[1] Long Short-Term Memory (LSTM) networks are a specialized class of Recurrent Neural Networks (RNNs) designed to capture long-range temporal dependencies, making them particularly effective for complex time series analysis and sequence prediction.

empirical testing, using the VitalDB dataset [5], demonstrates that synthetic sequences can be effectively generated by oversampling.

2 Methodology

2.1 Data and pre-processing

As mentioned in the Introduction, the time series representing the patient's blood pressure is coming from the VitalDB dataset [5]. This dataset is *"comprehensive of 6,388 surgical patients composed of intraoperative biosignals and clinical information. The biosignal data included in the dataset is high quality data such as 500 Hz waveform signals and numeric values at intervals of 1-7 seconds"*, as mentioned in the VitalDB description. The data was obtained from non-cardiac (general, thoracic, urologic, and gynaecological) surgery patients, categorised in: Demographic, Surgical approach, Anaesthesia, Duration of anaesthesia (min), Aesthetics and Device used during the medical operation. Even though this dataset is rich in information about patients, medical treatment and devices, only the mean blood pressure (MBP) read from the patient monitor (GE Solar 8000M) will be used to build a Machine Learning model capable of predicting hypotension events (MBP $<$ 65mmHg for at least 1 minute).

An example is given by the arterial MBP of patient-4841 (from VitalDB), shown in figure 1, where it is evident few MBP values are extremely high, probably due to brief monitoring issues or patient movement, as well as a few interruptions of the monitoring, probably due to temporary machine failure. Patient-4841 is one of the patient records available within the dataset, which is characterised by a high degree of heterogeneity in terms of both the presence of hypotensive events and the total duration of monitoring.

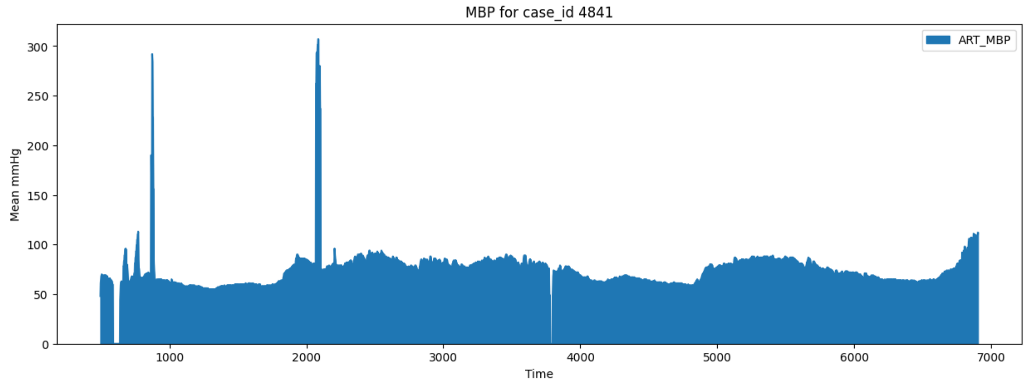

Figure 1 – VitalDB MBP time series for patient 4841, the x-axis is time in minutes and the y-axis is MBP in mmHg.

Before addressing the predictive modelling of hypotension, it is essential to implement a robust data-processing pipeline. An initial preprocessing phase is required to filter and adjust the patients' time series arterial MBP from the VitalDB dataset. For each patient, the following preprocessing steps are applied:

1. **Restriction of Cases**: Only patients over 18 years old, with case_id greater than 3600 and they are not transplant cases.
2. **Outlier Suppression**: MBP values exceeding 150 mmHg are replaced with NaN (Not a Number). This threshold is established because values above this level typically represent outliers or sensor errors rather than plausible physiological states in this clinical context.
3. **Missing Value Interpolation**: To maintain the continuity of the signal, linear interpolation is applied to fill NaN values. This addresses gaps caused by both the previous outlier removal step and intermittent sensor disconnections during monitoring.
4. **Boundary Truncation**: Any remaining NaN values, specifically those located at the extreme beginning or end of a time series where interpolation is not possible, are dropped to ensure a clean, continuous input for the model.

At this point, patients' MBP time series are restricted to 3420 cases, and the MBP are harmonised and ready for the next two steps of preprocessing: transformation and validation.

As mentioned in the Introduction, predicting hypotension events can be considered an event-detection task, or even more generally, a *Supervised Binary Classification Problem*. In fact, it could be possible to train a classifier capable of recognising if a certain input time window could later lead to an unexpected drops of blood pressure. Within this problem setting, the input data `X` to the classifier are sequences of MBP, and the output label `y` is the following time window where the hypotension episode occurs. However, two major changes are needed to this framework:

- To be clinically useful, an automated warning must provide a sufficient lead time for medical staff to implement corrective measures. Consequently, the predictive model cannot rely on data immediately adjacent to the event. A 'buffer' or gap period of several minutes must be maintained between the observation window (the input) and the prediction window (the output) to ensure that clinicians have a viable window for intervention before the onset of haemodynamic instability.
- In this framework, the model is not trained to characterise the duration or morphology of an active hypotensive event. Rather, the scope is restricted to the pre-hypotensive phase. By prioritising the likelihood of an upcoming crisis over the classification of an existing one, the study aims to provide actionable early warnings rather than redundant diagnostic labels for already manifested instability.

Based on these considerations, we reformulate the predictive framework by partitioning the MBP time series into distinct temporal segments. As illustrated in Figure 2, each segment is subdivided into three functional components: an observation window (`seq_x`), which serves as the model input; a waiting time interval (`black_out`), which simulates the necessary clinical gap for early warning; and a target window (`seq_y`), which contains the MBP data used to determine the presence or absence of a hypotensive event.

In other words, **segment** refers to the total MBP time window, while **sequence** is defined as any one of the three distinct sub-sections that constitute a single MBP segment.

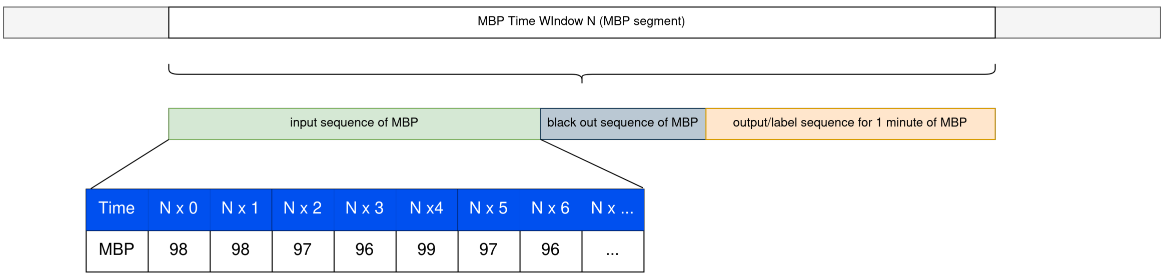

Figure 2 – VitalDB MBP time window partitioned in `seq_x` (in green), `black_out` (in black-gray) and `seq_y` (in orange)

However, how long should each segment of a time window? Each segment has the same number of MBP samples? For sure, the output sequence has to be at least 1 minute to determine if the disease is detected or not. In principle, the `black_out` sequence could be no less than 1 minute so that medical professionals have enough time to take the right actions in time, if necessary. Probably the most sensitive sequence is `seq_x`, which represents the input data to the LSTMClassifier: a sequence too short could not be enough to catch any underlying pattern in the data, resulting in a meaningless view of the patients' MBP, as shown in Figure 3. In fact, using a size of 10 MBP samples for `seq_x` results in an almost straight line, where the only takeaway would be the mean of the sequence. While the `seq_x` leading to hypotension is true to be below 65 mmHg, it is expected and therefore not particularly of interest.

The classifier, essentially, could only use the mean, and this is insufficient for achieving clear class separation. Given this, it is difficult to argue that data generation alone would improve the final task over standard balancing techniques. Increasing the amount of input information, such as enlarging the input window, appears far more promising. For example, with a sequence size of 300 MBP samples (10 minutes), it would be possible to consider not only the mean, but hopefully the seasonality and trend. As a matter of fact, figure 4 shows a representation of `seq_x` obtained by applying the UMAP technique, which extracts principal features for dimensionality reduction, as explained in [6].

The UMAP projections in Figure 4 reveal two distinct clusters that characterise the feature space of the input sequences. The blue cluster is notably compact and well-defined, suggesting a high degree of morphological

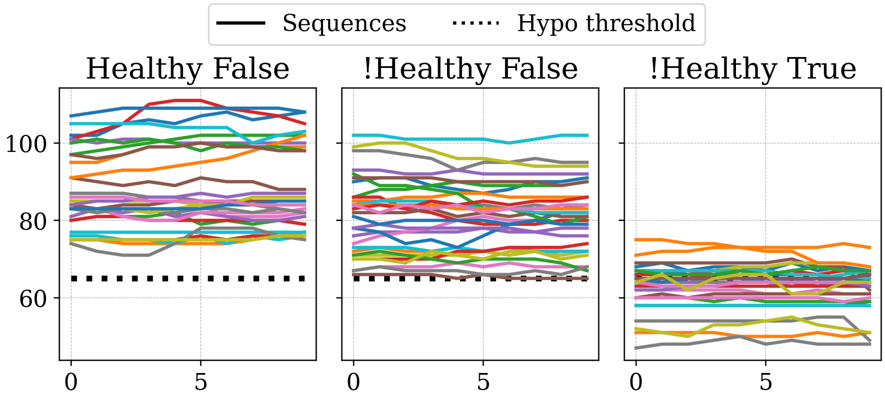

Figure 3 – Three subplots representing a subset of `seq_x` patients (of different colours), with 10 MBP samples. The subplots share the same axis, where the x-axis represents time and the y-axis represents the MBP value in mmHg; the dotted line represents the hypotension threshold of 65 mmHg. Leftmost subplot shows `seq_x` for healthy patient, centred subplot shows `seq_x` for not-healthy patient, but not leading to hypotension episode in `seq_y` sequence, and rightmost subplot shows `seq_x` for not-healthy patient leading to hypotension crisis in `seq_y`.

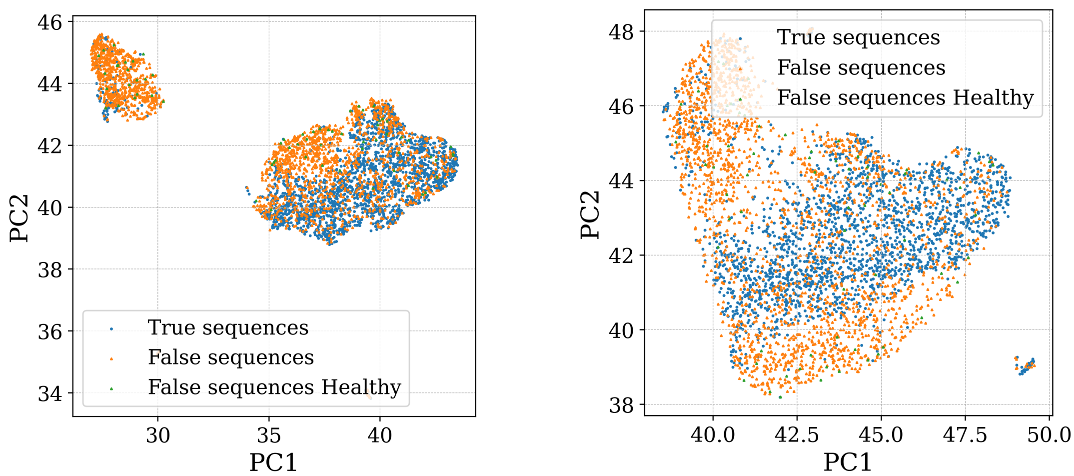

Figure 4 – UMAP visualisation of MBP sequences using different hyper-parameters. Both subplots use a pair of Principal Components on the axes extracted using UMAP. Both subplots present `seq_x` leading to hypotension (in blue) and sequences without the disease for healthy (green) and not healthy patients (orange).

similarity among sequences (seq_x) that immediately precede a hypotensive event. Conversely, the orange cluster is significantly more dispersed; this represents sequences that do not result in a blood pressure drop, despite belonging to patients with a history of haemodynamic instability. It is also worth noting that sequences from patients entirely unaffected by hypotension are relatively sparse, appearing as isolated points that lack the density to form a separated cluster. This distribution highlights the disproportionate ratio of stable to hypotensive subjects within this VitalDB cohort. Given that the number of patients afflicted by hypotension in this specific subset is minimal, these records may be excluded.

This transformation converts a patient's MBP from being a Time Series to an ordered collection of sequences, where the temporal order is preserved within each sequence and between consecutive sequences. In addition, a sequence size of 300 samples for seq_x has shown how a sufficiently long sequence can reveal more information than can be obtained from a shorter sequence.

The final step of the data preprocessing phase is the (sequences) validation: in this step, seq_x and seq_y are analysed from a clinical point of view; both sequences have to satisfy a set of constraints for the segment (i.e. a time window composed by seq_x, black_out and seq_y) to be considered valid. The medical constraints are:

- mean of the sequence should not be over 0.1 mmHg.
- None of the values in a sequence should be over 150 mmHg since, on average, it never exceeds this threshold under normal conditions.
- Considering the temporal order of the values within a sequence, none of the values should drastically change (in absolute value) between consecutive MBP values over 50 mmHg; in other terms, abrupt changes are not allowed within an MBP sequence.

Only for seq_y, an extra condition has to be satisfied: the maximum MBP value, within the sequence, should not be over 65 mmHg. If a sequence fulfils all the constraints, that is a valid sequence; otherwise, the sequence should be discarded. Finally, to align the new framework with the second consideration 2.1, it is enough to substitute the seq_y sequence with a binary value, which means True if the sequence seq_y presents hypotension and False otherwise. Something that was implicitly done during the previous analysis of the sequence seq_x.

Consequently, the MBP samples within seq_y are discarded in favour of this binary representation. Furthermore, for the sake of brevity in this report, the terms 'True seq_x' and 'False seq_x' are used to denote sequences that precede a positive (hypotensive) or negative (stable) seq_y, respectively (same approach is applied for seq_y).

This preprocessing stage ensures that the patient data adheres to the specific requirements of the chosen classification approach. However, significant inter-patient variability exists regarding both total monitoring duration and the frequency of hypotensive events: the number of sequences and segments varies between subjects, as does the frequency of hypotensive crises. Some patients present a high density of True seq_y labels, whereas others remain largely unaffected. Figure 5 depicts the resulting distribution of these labels among all patients, highlighting the inherent class imbalance present in the surgical monitoring data.

Furthermore, Figure 5 illustrates the distinct distributional characteristics of the dataset. The False seq_x sequences exhibit a Gaussian-like distribution, whereas the True seq_x sequences follow a geometric distribution with a significant positive skew. This left-ward concentration indicates that the majority of patients experience only a limited number of hypotensive events. Notably, the x-axis scales reveal a severe disparity in frequency magnitude: True cases are on the order of 10^3, while False cases reach 10^4. Although this is only a difference of ten, it is significant when it comes to frequencies.

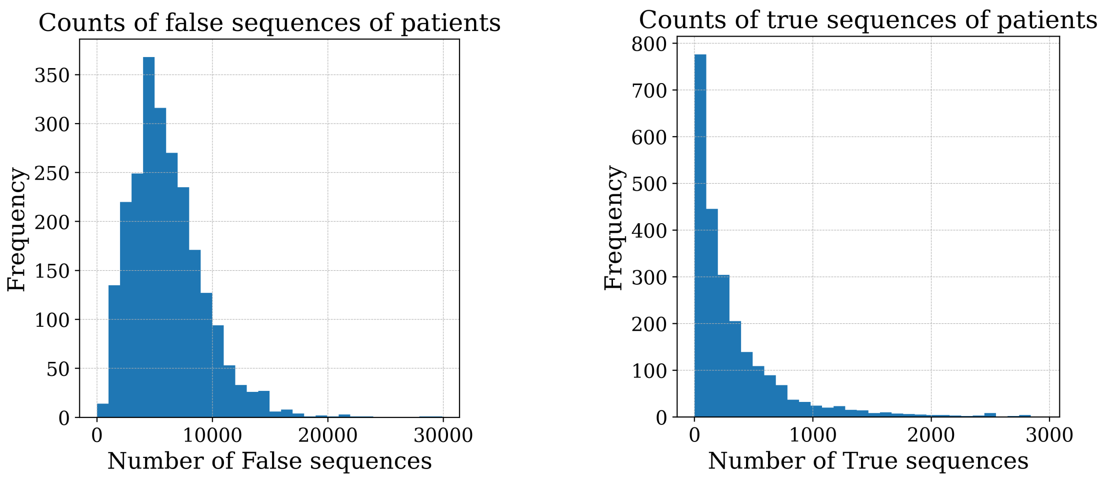

Figure 5 – Histograms for sequences `seq_x` leading to hypotension (True) and no hypotension (False).

2.2 Oversampling methods

The pronounced class imbalance, characterised by the dominance of non-event sequences, constitutes the central challenge of this case study. If left unaddressed, this disparity would likely result in a LSTMClassifier bias toward the majority class, undermining the classifier's ability to detect critical hypotension in advance. Consequently, it justifies the integration of oversampling strategies discussed in the subsequent sections. By balancing the dataset, we aim to mitigate the risk of majority-class bias and improve the reliability of our hypotension forecasts.

To address the severe minority class sparsity observed in the VitalDB dataset, a set of oversampling techniques is designed. These methods are categorised into three groups based on their underlying logic: Heuristic-based [H], Stochastic-noise [S], and Latent Space-based [L], listed as follows.

- **Random Oversampler** (`rover`) [H]: This method performs simple random sampling with replacement from the existing minority class. It increases the representation of hypotensive sequences by duplicating original observations. While it does not introduce new information, it effectively increases the weight of minority instances during the classifier training.

- **SMOTE** (`smote`) [H]: Synthetic Minority Over-sampling Technique (SMOTE) [4] generates synthetic samples by interpolating between existing minority instances. For a given True `seq_x`, it identifies its k-nearest neighbours [7] [8] and creates new points along the line sequences connecting them in the feature space, thereby expanding the decision boundary.

- **ADASYN** (`adasyn`) [H]: Adaptive Synthetic Sampling (ADASYN) [9] builds upon the SMOTE framework but uses a weighted distribution for different minority class examples according to their level of difficulty in learning. It generates more synthetic data for minority samples that are surrounded by majority class samples, focusing the model's attention on high-variance "borderline" regions.

- **Random Noise per Global Patient** (`rnoise_global`) [S]: This stochastic approach generates synthetic sequences by adding Gaussian noise ($\epsilon \sim N(0, \sigma_{global})$, to original sequences `seq_x` leading to the disease. Here, σ_{global} is the scaled standard deviation derived from the entire pool of available training sequences `seq_x` for a specific patient. It preserves the patient's global haemodynamic pattern while introducing subtle variations.

- **Random Noise Pooled** (`rnoise_pooled`) [S]: Similar to the global noise approach, this method utilises a local variance metric. The noise is sampled using a standard deviation (σ_{pooled}) [10] computed from the local variation within each specific True `seq_x` available per patient. This method ignores the fact that the mean is drifting and only focuses on the internal variance of each sequence; for k sequences, the pooled variance is computed as: $s^2_{pooled} = \frac{1}{k}\sum_i^k s_i^2$. This ensures that the synthetic data reflects the local short-term temporal volatility characteristic of that specific patient.

- **Random Noise Stationarized** (rnoise_stationarized) [S]: To capture the underlying dynamics of the time series without the influence of non-stationary trends (like gradual drifts in blood pressure), this method applies for each patient a first-order differentiation ($\Delta y_t = y_t - y_{t-1}$) to the ordered collection of sequences seq_x. The standard deviation is then calculated from this stationarized series. By adding noise based on the residuals of the differentiated series, the method synthesises data that respects the patient's underlying physiological "shocks" rather than their absolute pressure levels. By applying noise to the first-order differences, this technique focuses on the volatility of the MBP rather than its magnitude. This ensures that the synthetic data preserves the patient's unique physiological dynamics, such as sudden spikes or drops, without being influenced by the drift in absolute blood pressure levels.
- **Latent Diffusion Model** (ldm) [L]: Latent Diffusion Model [11] is a state-of-the-art generative approach that learns the underlying probability distribution of the seq_x sequences leading to hypotension. The model iteratively de-noises a Gaussian signal to synthesise highly realistic and diverse seq_x sequences. This specific use of ldm [2] focuses on each True seq_x, regardless of the temporal order between seq_x and without considering the general shape of the patient MBP time series. Unlike heuristic interpolation, the ldm can capture complex, non-linear dependencies in the MBP time series, providing the most sophisticated form of data augmentation.

Because the rover method simply duplicates existing observations rather than expanding the feature space with new synthetic samples, it lacks the capacity to introduce additional feature diversity. Consequently, it is established as the **baseline** oversampling benchmark to determine if more complex oversampling methods provide a significant performance gain.

2.3 Infrastructure and framework

The experimentation is orchestrated using MLFlow [12], an open-source platform designed to manage the end-to-end machine learning lifecycle. It provides essential tracking of hyperparameters, code versions, and metrics, ensuring that every oversampling trial is reproducible and comparable.

The entire code infrastructure is based on six sequential pipelines, which ensure a modular, clear separation of scopes and computational efficiency:

1. **Data & Transformation Pipelines**: Handle the initial ingestion and feature engineering of the Mean Blood Pressure (MBP) time series coming from the VitalDB dataset and previously explained.
2. **K-Fold Pipeline**: To optimise resource consumption, the dataset is partitioned into 10 folds prior to the Cross-Validation, and each fold holds train and test subsets. By persisting these folds in storage, we eliminate the redundant overhead of re-partitioning for every oversampling strategy, significantly reducing computational latency and I/O operations.
3. **Cross-Validation (CV) Pipeline**: This serves as the core execution engine. The infrastructure employs an *Adapter Pattern*, where an abstract ClassBalancer base class is extended into specific Adapter implementations for each oversampling method (e.g., rover, smote). The CV pipeline retrieves a pre-computed fold from the previous pipeline, passes the training set to the active Adapter for balancing, and ensures the test set remains strictly untouched for unbiased evaluation.
4. **ML Pipeline**: Receives the augmented training data to fit the LSTMClassifier. We assess the classifier using standard metrics, including Accuracy, Precision, F1, and F2. However, due to the severe class imbalance and the critical nature of patient monitoring, Accuracy and Precision are considered secondary. Instead, we focus on Recall to measure the detection rate of pre-hypotensive sequences and F2 score to provide a weighted evaluation that favours sensitivity over precision.

[2] The employed Latent Diffusion Model is based on the SECURED project and package SDG_CTG; it is equipped exclusively with the Identity technique, and it is simplified to a single channel (instead of two channels) since the input data is one-dimensional.

5. **Summary Pipeline**: Acts as the aggregation step, assembling the cross-validation results across all folds and oversampling strategies to facilitate a comparative analysis.

Except for `Summary Pipeline`, which wraps all partial results together, a high overview of the infrastructure is shown in figure **??**, where the previous pipelines are presented and the way they are connected.

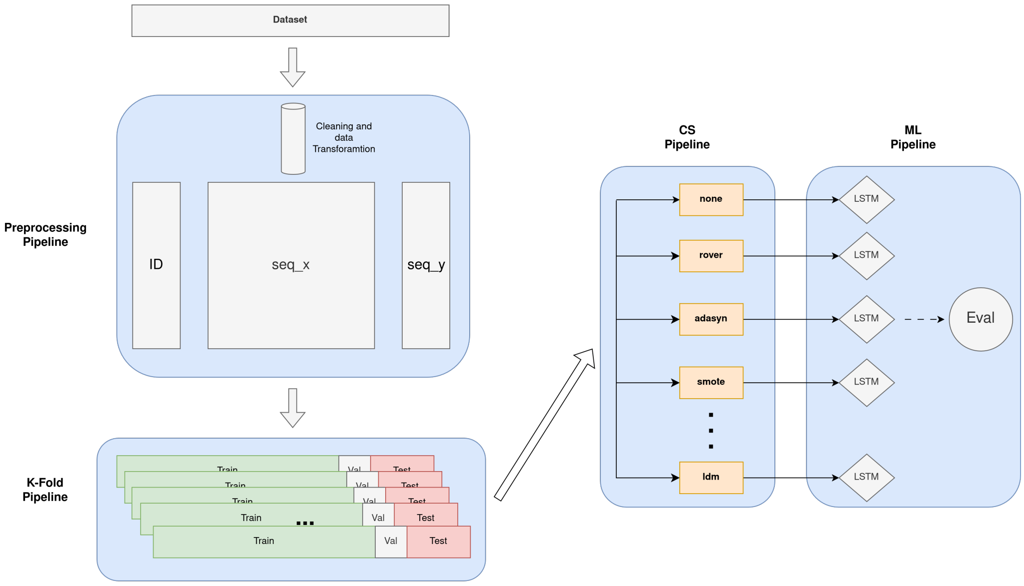

Figure 6 – Experiment infrastructure with main data pipelines.

As established in the Introduction, an LSTM-based architecture is employed to address the event-detection task. However, in the context of this study, the classifier does not serve as the primary object of optimisation; rather, it functions as a standardised validator to evaluate the efficacy of various minority-class oversampling strategies.

To ensure a fair comparison among the oversampling methods, the LSTM architecture is kept static throughout all experiments. The model is composed of:

- An Input Layer configured for the temporal dimensions of `seq_x`.
- A Hidden LSTM Layer with 16 units to extract temporal features.
- A Batch Normalisation Layer [13] to stabilise the learning process and mitigate internal covariate shift.
- A Dense Output Layer with a single neuron and a Sigmoid activation function.

Since the Sigmoid function outputs a probability P(hypotension`seq_x`), a decision threshold of 0.5 is applied to generate the final binary classification. The model was trained with 100 epochs, and it is optimised using the Adam optimiser and the Binary Cross-Entropy (BCE) loss function.

The Cross-Validation (CV) process is specifically designed to assess the external impact of the oversampling methods rather than the intrinsic hyperparameters of the LSTM itself. By utilising the classifier as a fixed "Validator", it is possible to isolate and quantify the performance shifts directly attributable to the synthetic data generator and its underlying oversampling technique.

As mentioned in the Introduction, the classifier evaluation is performed with several evaluation criteria, notably the Accuracy, Precision, Recall, F1 and finally the F2; all well-known metrics and widely used in Classification Problems, except for F2, are not particularly standard. F2 score is a variant of F1 score where the trade-off between Precision and Recall is more in favour of the latter, as depicted by the formula:

$$F_\beta = (1+\beta^2) \times \frac{Precision \times Recall}{\beta^2 \times Precision + Recall}$$

Where $\beta = 2$ in this case. This approach is essential in healthcare scenarios where failing to detect a critical event (a False Negative) is significantly more costly than a false alarm.

method	recall_score	f1_score	f2_score	precision_score	accuracy_score
adasyn	0.817	0.319	0.461	0.217	0.87
rnoise_global	0.9	0.364	0.521	0.253	0.875
rnoise_pooled	0.817	0.327	0.47	0.226	0.88
rnoise_stationarized	0.811	0.329	0.469	0.227	0.88
ldm	0.293	0.273	0.273	0.33	0.94
rover	0.782	0.344	0.48	0.244	0.889
smote	0.765	0.338	0.469	0.24	0.89
none	0.223	0.253	0.229	0.442	0.95

Table 4 – The main evaluation metrics in the medical field used to compare multiple oversampling methods.

3 Results

The results of the conducted experiments are shown in Figure 7, where the 10-Fold CV evaluation outcome is illustrated with a bar-chart with 95% confidence interval, for each oversampling method.

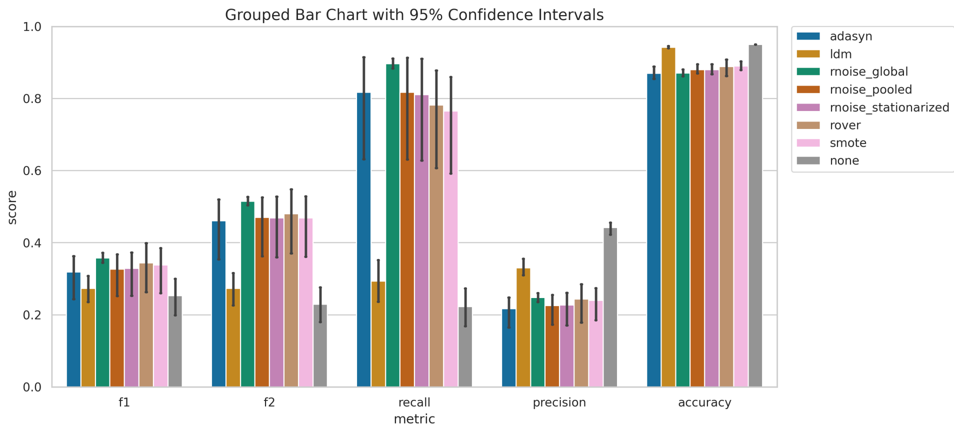

Figure 7 – Grouped Bar chart for oversampling methods with 10-Fold Cross Validation reduced to a 95% confidence interval. On the x-axis, the evaluation metrics for the LSTMClassifier (F1, F2, Recall, etc.), on the y-axis, the metrics score ranging from 0 to 1.

Figure 7 provides a qualitative overview of the evaluated oversampling strategies. Each bar represents the mean performance metric across the 10-Fold Cross-Validation, with error bars indicating the associated confidence intervals.

A primary observation is that the non-augmented baseline (`none`) yields poor performance, particularly in terms of Recall, which is the metric directly targeting the hypothension. Furthermore, most oversampling techniques exhibit a confidence interval of approximately ±8% for the F1 and F2 scores. This variability is even more pronounced for Recall, where fluctuations reach up to 12%, suggesting that the detection of rare hypotensive events is highly sensitive to the specific set of patients present in each fold. In contrast, the confidence intervals for Precision and Accuracy are notably narrow, particularly for Accuracy. This lack of variability implies that these metrics are relatively indifferent to the specific oversampling intervention, further reinforcing the suspicion that they are being driven by the dominant majority class.

Despite the narrow intervals, the high Accuracy and Precision scores for the non-balanced method warrant a deeper critical analysis. These figures appear to be statistically biased by the class imbalance rather than a reflection of true predictive power.

A detailed numerical breakdown of these findings is provided in Table 4, which summarises the mean scores across all tested methodologies.

An initial comment is for the `none` method (i.e., not applying any balancing approach). While most performance metrics plateau around 25%, the Accuracy reaches a deceptive 95%. This discrepancy is a classic manifestation of the Accuracy Paradox [14], where the classifier achieves a high success rate simply by predicting the majority class (non-hypotension). Furthermore, the relatively higher Precision (44%) compared to other metrics is equally misleading; it suggests that while the model is cautious, it is fundamentally failing to capture the minority class, rendering it clinically ineffective for early warning systems. More to the point, Precision's conclusion is 'unmasked' by the F1 that factors in both classes, resulting in the lowest score among all the metrics.

Taking into account `rover` is the baseline strategy, and by looking at the categorical groups of the oversampling methods (i.e. Heuristic-based [H], Stochastic-noise [S], and Latent Space-based [L]), it is interesting to observe how the Heuristic-based methods perform worse than the baseline in all evaluation metrics.

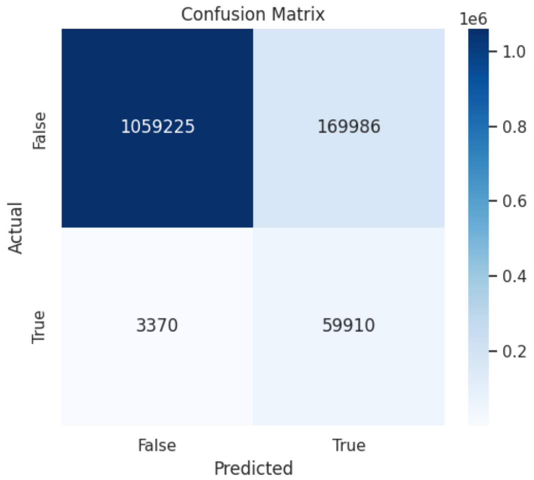

Figure 8 – `rnoise_global` Confusion Matrix related to the fourth Cross-Validation fold.

The Latent Diffusion Model (`ldm`) underperformed significantly, failing to surpass the baseline "no-balancing" method across most evaluation metrics. This subpar performance may be attributed to several architectural and procedural factors rather than the inherent capability of the model.

As the `ldm` was implemented "as-is" from the SECURED project without domain-specific fine-tuning, the current configuration may not be optimised for the specific nuances of this dataset. Potential bottlenecks include:

- Errors in the Variational Autoencoder (VAE) training phase may have introduced information loss during the latent mapping process.
- The choice of sampling strategy and a potentially insufficient number of sampling steps likely prevented the model from reaching a high-fidelity data distribution.

To improve the utility of generative diffusion for this use case, the following avenues should be explored:

- Conduct a grid search on the DDIM sampler settings, specifically tuning the η (stochasticity) parameter to better capture data variance.
- Evaluate different latent representations beyond the current Identity technique and experiment with alternative samplers (e.g., DPM-Solver) to improve convergence.
- The current hypotension sequences may be too short for the `ldm` to learn meaningful temporal dependencies. Increasing the sequence window could provide the context necessary for more accurate generation.
- Rather than localised generation, the `ldm` could be trained on the entire patient time-series data. By generating complete synthetic patients and then extracting specific sequences associated with hypotension, we may achieve a more robust and balanced dataset.

A different behaviour is shown from the Stochastic-noise methods, which are performing equally to the baseline or even better. Notably, the `rnoise_global` method reaches 90% in Recall, an extremely high score. Similar for F2, `rnoise_global` is the method with the highest score of 52%; even for the F1 score, which is balancing Recall and Precision, this Stochastic-noise method is the best.

Similarly, Figure 8 displays the confusion matrix for `rnoise_global` during the fourth Cross-Validation fold. Notably, the count of False Negatives (instances where impending hypotension was erroneously classified as stable) is at its absolute minimum. This outcome validates the efficacy of the oversampling strategy in minimising the clinical risk associated with undetected haemodynamic crises. However, an analysis of the confusion matrix also reveals a significant limitation: the number of False Positives remains notably high. This represents the inherent cost of prioritising Recall; while the model successfully avoids missing critical events, it does so at the expense of Precision.

A different perspective of the results is given by the table 7, where only `none`, `rover` (baseline) and `rnoise_global` methods are compared, to understand how much the gain (and loss) of the best oversampling method is.

	%_recall_score	%_f1_score	%_f2_score	%_precision_score	%_accuracy_score
rnoise_global vs none	+67.7 ↑	+11.1 ↑	+29.2 ↑	-18.9 ↓	-7.5 ↓
rnoise_global vs rover	+11.8 ↑	+2 ↑	+4.1 ↑	+0.9 ↑	-1.4 ↓

Table 5 – The best oversampling method is compared to no method applied and random oversampling, showing the gain and loss in percentage.

The comparative analysis in Table 7 demonstrates that `rnoise_global` significantly outperforms not balancing the dataset, most notably in Recall, where it achieves a substantial gain of 67.7%. Furthermore, it maintains a performance lead over the `rover` benchmark, exceeding its Recall by nearly 12%. While there is a marginal decrease in Precision and Accuracy, this represents a strategically acceptable trade-off in a clinical context: prioritising Recall ensures the identification of high-risk patients, and with Accuracy sustained at a robust average of 80%, the model's overall reliability remains high.

Another interesting aspect of the experiment is shown in Figure 9, which compares execution times.

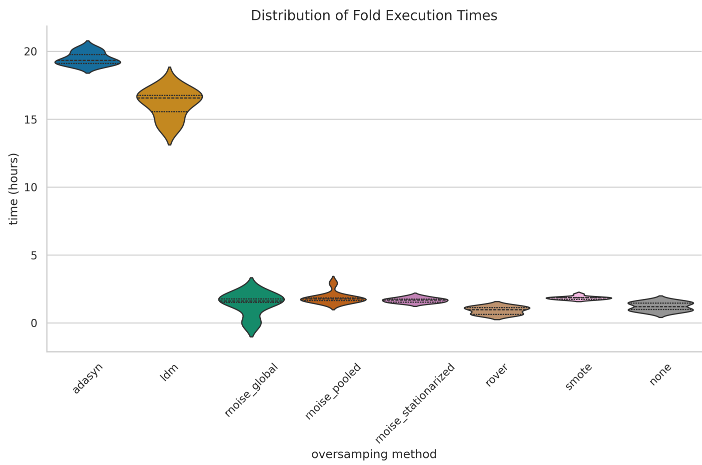

Figure 9 – Distribution of 10-Fold CV execution time for each balancing approach.

Figure **??** illustrates the execution time for the 10-Fold Cross-Validation across all strategies using violin plots. The distributions are generally narrow and uniform, with most methodologies converging at an average execution time of approximately 3 hours. A significant outlier is observed with the `adasyn` method, which required nearly 20 hours to complete the dataset balancing and training cycle. This exceptional computational overhead is likely attributable to adasyn's density-based mechanism: unlike simpler heuristic methods, `adasyn` adaptively generates samples based on the local distribution of the minority class, which may increase algorithmic complexity when processing high-dimensional MBP time series.

Similarly, the `ldm` exhibited execution times comparable to the `adasyn` method. This result is expected given the LDM's higher architectural complexity, which typically necessitates significantly greater computational overhead during both the denoising process and latent space transformations.

While the primary objective of this study is predictive performance rather than computational optimisation, these results provide a crucial secondary metric for assessing the scalability and practical feasibility of these models in clinical deployment.

Ultimately, the integration of oversampling, specifically the `rnoise_global` variant, is essential to mitigate majority-class bias and provides the most favourable metric profile for this case study.

4 Conclusion

This study investigated the impact of various oversampling strategies on the detection of hypotensive events within the highly imbalanced VitalDB dataset. While traditional and advanced methods marginally improved the classification correctness, only the Stochastic-Noise approach proved to be most effective: oversampling methods belonging to this category are based on a normal distribution, where the standard deviation is calculated dynamically from the patient's actual MBP monitoring data.

After several experiments, it was proven that the Stochastic-Noise approach `rnoise_global` is the best oversampling method, as shown in Table 6, where the augmented dataset is evaluated using an LSTMClassifier (variant of LSTM) using *10-Fold Cross Validation*.

method	recall_score	f1_score	f2_score	precision_score	accuracy_score
rnoise_global	0.9	0.364	0.521	0.253	0.875
rover	0.782	0.344	0.48	0.244	0.889
none	0.223	0.253	0.229	0.442	0.95

Table 6 – The main evaluation metrics in the medical field used to compare multiple oversampling methods. The table shows only the evaluation for `rnoise_global`, `rover` and `none`, which is the original dataset without any augmentation applied.

Table 6 provides a representative subset of the experimental results, focusing on the critical benchmarks of the study: the baseline duplication strategy (`rover`), the non-augmented control (`none`), and the top-performing stochastic method (`rnoise_global`).

The control strategy (`none`) consistently underperforms across all primary metrics, with the exception of Accuracy. This discrepancy is a classic manifestation of the Accuracy Paradox in imbalanced datasets [14]: the model achieves high global accuracy by over-optimising for the majority (stable) class while failing to identify hypotensive events. In contrast, the `rnoise_global` method emerges as the best balancing strategy, achieving a Recall of 90% and an F2 score of 52%. While the F2 score reflects a trade-off between sensitivity and precision, the high Recall demonstrates the model's prioritised capability to ensure patient safety by minimising missed events.

These results indicate that by introducing global Gaussian noise, the LSTMClassifier effectively generalise on the minority class and, consequently, captures rare pre-hypotensive patterns. Table 7 further quantifies the relative performance gain achieved by integrating this synthetic data generation approach into the training pipeline.

	%_recall_score	%_f1_score	%_f2_score	%_precision_score	%_accuracy_score
rnoise_global vs none	+67.7 ↑	+11.1 ↑	+29.2 ↑	-18.9 ↓	-7.5 ↓
rnoise_global vs rover	+11.8 ↑	+2 ↑	+4.1 ↑	+0.9 ↑	-1.4 ↓

Table 7 – The best oversampling method is compared to no method applied and random oversampling, showing the gain and loss in percentage.

Both 6 and 7 tables detail the Precision and Accuracy scores, which exhibit a relative decline compared to the unbalanced and random oversampling methods. This suggests a performance trade-off, where the focus on improving minority class representation slightly reduces the LSTM's global accuracy and precision.

In conclusion, this study shows how a Synthetic Data Generator for medical time series can bring significant benefits for rare events prediction. In the clinical context of VitalDB, where a missed detection is a missed opportunity to save a life, an intelligent oversampling represents a key factor for an effective prevention of hypotension events.

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